MIL-I-27191B(USAF)
1 NOVEMBER 1979
SUPERSEDING
MIL-I-27191A (USAF)
2 November 1965

MILITARY SPECIFICATION

INDICATOR, POSITION, VARIABLE NOZZLE

This specification is approved for use by Code 99 Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for 2 types of 28V dc, integrally lighted, variable nozzle position indicators.

1.2 Classification. The indicators shall be classed as follows:

MS24572-3	Red integral lighting
MS24572-4	White integral lighting

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6620

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3.4.1.2 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±5°C for a minimum of 2 hours and then the face of the glass immediately immersed in an ice water bath for 3 minutes.

3.4.2 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint - 65° C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. Other concentrations of helium shall not be used except with prior approval of the procuring activity. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.4.3 Cover glass position. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.5 Dial. The dial shall be as shown in MS24572. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. The diameter of the dial shall be not less than 1-3/4 inches when measured across the outside ends of the graduations.

3.5.1 Markings. The finish and dimensions of the dial markings shall be as follows:

Markings	Height or length Inches ±0,016	Width of Line or graduation Inches ±0.005	Color FED-STD-595
Numerals 0,20,40, 60,80,100	0.125	0.016	No. 37875
Major graduations	0.141	0.025	No. 37875
Minor graduations	0.093	0.016	No. 37875
Lettering "NOZZLE"	0.125	0.020	NO. 37875
Lettering PERCENT"	0.0625	0.010	No. 37875
Lettering "OPEN"	0.093	0.016	No. 37875
Lettering "MIN'	0.093	0.016	No. 37875
Lettering "MAX"	0.093	0.016	No. 37875
Lettering "MS24572 (3 or 4 as applicable)	0.063	0.010	No 37038

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3.6 Pointer. The pointer shall conform essentially to MS33585-8 and shall be suitable for integral lighting. The pointer hub shall not obscure the marking "Percent" on the dial.

3.6.1 Pointer dampening. The pointer shall be sufficiently dampened to prevent an oscillations in excess of ±1.0 percent while operating and while the transmitter is rotated in short rapid increments. The pointer shall dampen to zero motion within a time period of 1.0 ±0.25 second after it is given a step increment of 65 percent.

3.7 Electrical connector. Unless otherwise specified, the electrical connector shall be a Bendix Type PT-IH-10-10-6P or equal.

3.8 Integral lighting. The indicator covered by this specification shall be integrally lighted in accordance with the specifications indicated below except as otherwise specified:

MS24572-3 - MIL-L-25467
MS24572-4 - MIL-L-27160

3.8.1 Lamp. The light source for the indicator shall be three or more MS25237 or MS24515 lamps.

3.8.1.1 Lamp position. The lamps shall be located within the hermetically sealed inclosure.

3.8.1.2 Lamp replacement. The lamp installation shall be designed so that replacement lamps can be accurately positioned when replacement is necessary. Replacement will be accomplished at overhaul depots.

3.9 External wiring. The external wiring shall be a shown on figure 1.

3.10 Identifiction of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

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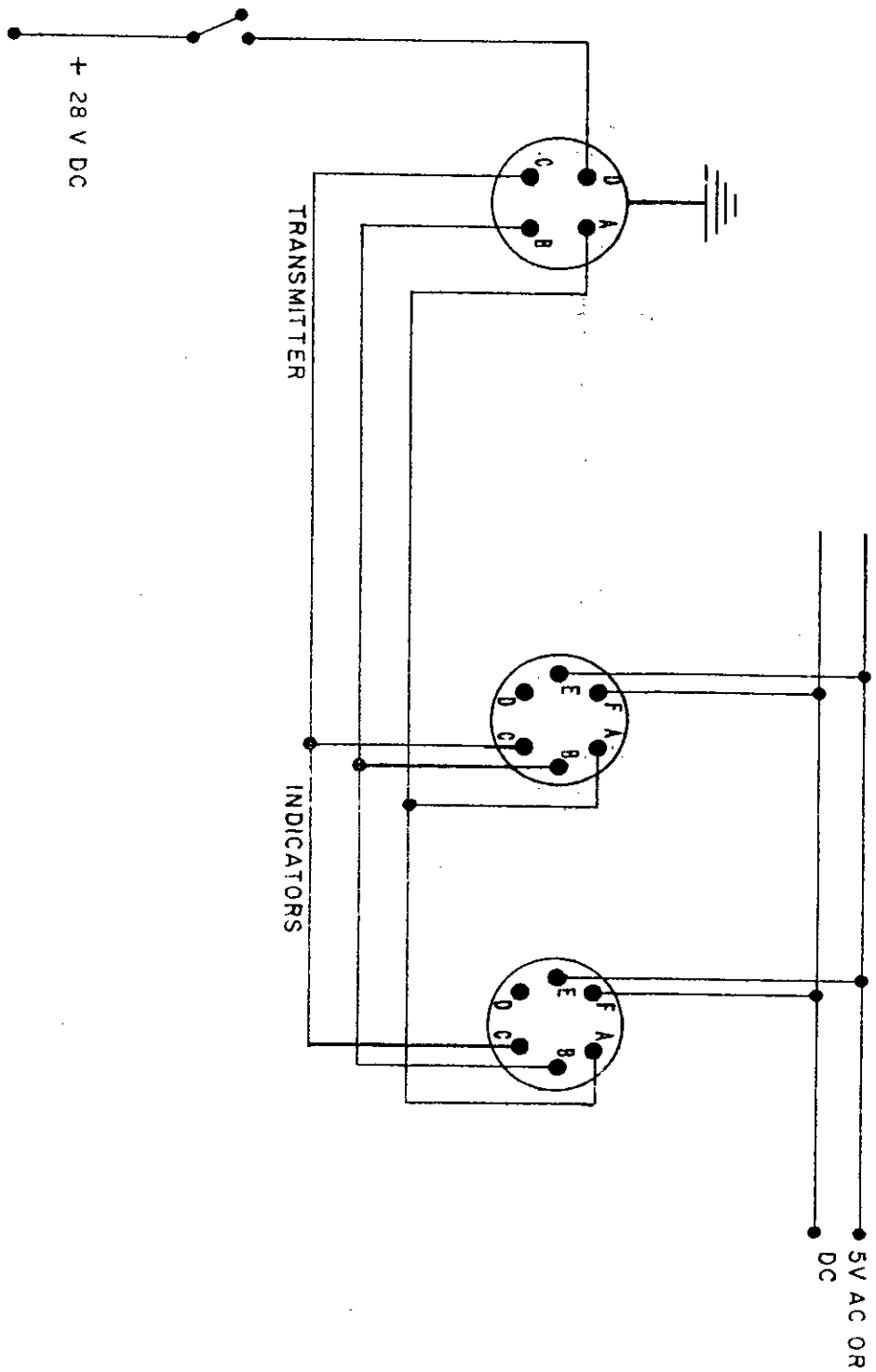


FIGURE 1. External Wiring Diagram

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Sampling, inspection, and tests shall be as specified in MIL-I-7062, except as specified herein.

4.3 Test conditions

4.3.1 For the purpose of this specification the identification of the tests shall be as follows:

MIL-I-27191	MIL-I-7062
Acceptance tests	Inspection tests
Sampling plan A	Sampling tests
Sampling plan B	Special tests
Qualification tests	Qualification tests

4.3.2 Scale error test. The scale error shall be read from the test transmitter scale with the indicator pointer on each graduation of the indicator dial.

4.3.2.1 A scale error test shall be conducted using two indicators with one master transmitter. The accuracy of each indicator during this test shall not exceed 1.25 times the single indicator accuracy specified.

4.3.2.2 Accuracy. The accuracy for a single indicator shall be as follows:

a. ±5 mechanical degrees for room and high temperature tests:

b. ±6 mechanical degrees for vibration and low temperature tests

4.4 Acceptance tests. Acceptance tests shall consist of individual tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

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4.4.1 Individual tests. The indicator shall be subjected to and meet the individual tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.2 Sampling plan A

4.4.2.1 The indicator shall be subjected to and meet the sampling plan A tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.2.2 Vibration error. The indicator shall be mounted on vibration stand in its normal operating position. The indicator shall be electrically connected for operational and lighting conditions. The indicator shall be subjected to vibration at an amplitude setting between 0.009 and 0.011 inch at frequencies varying from 300 to 3000 cpm. During vibration maximum amplitude of oscillation and variation of the movable pointer shall not exceed 1/32 inch. All of the lamps shall be operating at the completion of this test.

4.4.2.3 Pointer dampening. The indicator shall be electrically connected for operation, and the transmitter shall be rotated in short rapid increments. The amount of pointer oscillation shall not exceed ±1.0 percent. The indicator shall then be connected to two transmitters which shall be electrically separated by a switch. One transmitter shall be set to read 20 percent and the other transmitter shall be set to read 85 percent. The indicator connections shall be switched from one transmitter to the other (by use of the switch). The time interval from initiation of the switch to the instant the pointer comes to a complete stop shall be measured and shall not exceed 1.0 ±0.25 second. A stopwatch may be used to measure the time interval

4.4.3 Sampling plan B

4.4.3.1 The indicator shall be subjectd to and meet the sampling plan B tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.3.2 Vibration failure. The indicator shall be subjected to the vibration test, procedure IV of MIL-E-5272 for a period of 3 hours. The test shall be conducted while the indicator is being operated on 28V dc and while the bulbs are illuminated with 5V power. The indicator shall be subjected to vibration with an amplitude between 0.018 and 0.020 inch. After

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vibration, the indictor shall be subjected to room temperature operation and position and error tests. In addition, no looseness in the mechanism nor damage to any part of the indicator shall result from the vibration.

4.4.3.3 Fogging. The indicator shall be placed in a controlled ambient temperature of 85 ±5°C for a minimum of 2 hours. Within 10 seconds after removal from the oven, the indicator face shall be immersed to a depth of 1/4 to 1/3 inch in an ice water bath maintained at 0° to 5°C. After 3 minutes the indicator shall be removed from the ice bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected for evidence of fogging or damage. Evidence of fogging or damage shall be cause for rejection.

4.5 Qualification testing

4.5.1 Qualification test samples. The qualification test samples shall consist of three indicators, samples shall be identified with the manufacturer's own part number and any additional information required by the procuring activity.

4.5.2 Qualification tests. The qualification tests shall consist of all the tests specified herein.

4.6 Packaging Inspection.

4.6.1 Quality Conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.6.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

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5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with method 1C-1 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.6.2 of this specification.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping container shall conform to specification PPP-B-636, class weather resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

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5.3 Marking. In addition to any special marking required by the contract or order, each unit and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The variable nozzle position indicator covered by this specification is intended to be used for indicating the position of variable exhaust nozzles, on turbo jet engines so equipped, in percent of maximum area. With this indicaton, the pilot is able, with proper interpretation, to infer the performance of the engines afterburner complex.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Level of packaging and packing desired
c. Whether sampling plan B tests are to be conducted.

6.3 Qualification With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manfacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is HQ AFLC CASO/LODS, Federal Center, Battle Creek MI, 49016 and information pertaining to qualification of products may be obtained from that activity.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 99

Project Number:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-27191B(USAF) INDICATOR, POSITION, VARIABLE NOZZLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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